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By

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DECLARATION

I declare that the project entitled "*Raj's Store — AI-Powered E-Commerce Web Application*" has been prepared by me under the guidance of "Dr. Parul Saini". No part of this project has formed the basis for the award of any degree or fellowship previously.

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CERTIFICATE

This is to certify that the project entitled "Raj's Store — AI-Powered E-Commerce Web Application" has been carried out by Abhay Raj Singh bearing Roll No. CS-2341385 in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering from IILM University, Greater Noida, during the academic year 2025–2026.

The project work is original and has not been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

Raj's Store is a fully functional, AI-powered e-commerce web application developed using core web technologies — HTML5, CSS3, and Vanilla JavaScript. The application simulates a premium online retail experience with a complete product catalogue spanning fashion, electronics, and lifestyle categories. Key features include a login/authentication modal, a dynamic product grid with search and sort functionality, an interactive shopping cart with quantity controls, a Wishlist system, a quick-view modal for products, real-time toast notifications, and a conversational AI-powered personal shopping assistant chatbot.

The chatbot is integrated as an intelligent rule-based engine acting as a personal stylist, providing personalised product recommendations based on user budget, occasion, and preferences. The application is entirely client-side, requiring no backend infrastructure, making it lightweight, portable, and deployable on any static hosting platform.

The project demonstrates key concepts in front-end development including DOM manipulation, event-driven programming, responsive UI design, API integration, and user experience design. The application is designed with a consistent purple-gradient design language and supports mobile-responsive layouts.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
DOM	Document Object Model
HTML	HyperText Markup Language
JS	JavaScript
JSON	JavaScript Object Notation
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator
HTTP	HyperText Transfer Protocol
NLP	Natural Language Processing

CHAPTER-1

INTRODUCTION

1.1 Overview

Raj's Store is a client-side e-commerce web application developed using HTML5, CSS3, and Vanilla JavaScript. It simulates a real-world premium online retail experience without requiring any backend server or database. The application is designed to be lightweight, fast, and accessible across devices.

The project was conceived to demonstrate comprehensive front-end development skills including dynamic DOM manipulation, state management in JavaScript, responsive design, third-party API integration, and user experience engineering.

1.2 Motivation

The rapid growth of e-commerce in India — driven by platforms such as Flipkart, Amazon, and Meesho — highlights the need for well-designed, high-performance shopping interfaces. This project serves as a practical implementation of such an interface, showcasing how a rich shopping experience can be delivered entirely on the client side.

The integration of an AI-powered personal shopping assistant reflects the industry trend of embedding conversational AI into retail applications to enhance personalisation and customer engagement.

1.3 Objectives

- To design and develop a fully functional e-commerce front-end without any backend infrastructure.
- To implement core shopping features: product listing, search, sort, cart, and Wishlist.
- To integrate a free, zero-configuration AI chatbot that provides personalised product recommendations.
- To ensure a responsive, visually consistent design across desktop and mobile devices.
- To demonstrate modular JavaScript architecture through clean separation of concerns.

1.4 Scope

The scope of this project includes the complete front-end of an e-commerce application with six products across three categories, an AI shopping assistant, user authentication simulation, and a checkout demonstration. The application does not include payment gateway integration, real user accounts, or server-side logic, as these are outside the scope of this academic project.

1.5 Organisation of the Report

Chapter 2 presents a literature study of related web technologies and e-commerce platforms. Chapter 3 describes the dataset (product catalogue) and its structure. Chapter 4 covers the methodology and novelty of the project. Chapter 5 discusses experimental results and UI testing. Chapter 6 presents conclusions and future directions.

CHAPTER-2

LITERATURE STUDY

2.1 E-Commerce Web Applications

E-commerce web applications have evolved significantly since the early 2000s, transitioning from simple HTML pages to complex, JavaScript-heavy single-page applications (SPAs). Modern e-commerce sites such as Amazon and Flipkart rely on frameworks like React and Angular for their front-end layers, enabling dynamic rendering and state management at scale.

Several studies have examined the impact of UI/UX design on conversion rates in e-commerce. Nielsen (2012) found that clear navigation, fast load times, and intuitive product search are the top factors influencing purchase decisions. These findings informed the design choices in Raj's Store.

2.2 JavaScript and DOM Manipulation

Vanilla JavaScript, without any framework, remains a powerful tool for building interactive web applications. The Document Object Model (DOM) API provides a programmatic interface to HTML, enabling dynamic content updates, event handling, and animation — all heavily used in this project.

This project uses ES6+ features including arrow functions, template literals, destructuring, spread operators, and the Fetch API, demonstrating proficiency in modern JavaScript without framework dependency.

2.3 AI Chatbots in Retail

The deployment of conversational AI in e-commerce has grown substantially. Tools like Amazon's Alexa, Shopify's Kit, and various GPT-4 powered assistants demonstrate the value of natural language interfaces in guiding customers through product discovery.

For lightweight deployments, rule-based chatbots provide effective, zero-cost solutions. This project employs a keyword-matching rule engine designed to be API-swappable for future LLM integration.

2.4 CSS Design Trends

Modern CSS features including Flexbox, Grid, CSS custom properties, and keyframe animations have made it possible to build sophisticated, responsive layouts without JavaScript. The purple gradient design language of Raj's Store is implemented entirely through CSS, leveraging linear-gradient, backdrop-filter, and animation properties.

2.5 Summary

The literature review confirms that client-side e-commerce applications are both technically feasible and commercially relevant. The technologies selected — HTML5, CSS3, and Vanilla JavaScript — represent industry-standard tools that provide the full capability required for implementing the specified features.

CHAPTER-3

DATASET PREPARATION AND PRE-PROCESSING

3.1 Product Catalogue Design

The product catalogue for Raj's Store was manually curated to represent a diverse set of product categories typical of an Indian e-commerce platform. The catalogue comprises six products spanning three categories: Fashion, Electronics, and Lifestyle. Each product is a JavaScript object with the following attributes:

- `id` — Unique integer identifier.
- `name` — Product display name.
- `price` — Price in Indian Rupees (integer).
- `img` — URL to a publicly accessible product image (Picsum Photos API).
- `category` — Category string: 'fashion', 'electronics', or 'lifestyle'.
- `description` — Brief marketing description of the product.

3.2 Product Data

1. Violet Dream Hoodie — ₹2,499 | Fashion | Premium oversized hoodie with soft fleece lining.
2. Amethyst Wireless Earbuds — ₹3,299 | Electronics | High-fidelity sound, ANC, 30-hr battery.
3. Lunar Purple Sneakers — ₹3,899 | Fashion | Stylish sneakers with premium cushioning.
4. Crystal Glow Smartwatch — ₹5,299 | Electronics | Fitness tracker with AMOLED display.
5. Midnight Velvet Jacket — ₹4,199 | Fashion | Elegant velvet jacket with premium finish.
6. Aura LED Lamp — ₹1,799 | Lifestyle | Smart mood lighting with 16 million colours.

3.3 Image Pre-Processing

Product images are sourced dynamically from the Picsum Photos API (<https://picsum.photos>), which provides high-quality placeholder images at specified dimensions. Images are fetched at 600×600 pixels and rendered within product cards using CSS `object-fit: cover` to maintain aspect ratio. No offline image processing was required.

3.4 Data Validation

Client-side validation is applied through the `filterProducts()` function, which sanitises user search input before applying it to the product array. The sort logic validates the selected sort value against known options before executing array sort operations, preventing rendering errors.

CHAPTER-4

METHODOLOGY & NOVELTY

4.1 System Architecture

Raj's Store follows a three-layer client-side architecture:

- **Presentation Layer** — HTML5 provides the semantic structure of all UI components including the navbar, product grid, sidebars, and modals.
- **Style Layer** — CSS3 defines all visual styling, layout (Flexbox/Grid), animations, and responsive breakpoints.
- **Logic Layer** — Vanilla JavaScript handles application state (cart, wishlist), dynamic rendering, event handling, and chatbot logic.

4.2 Core Feature Implementation

Authentication: The login modal captures email and password inputs. Upon submission the modal is hidden and the main store content is revealed, simulating authentication without a backend.

Product Grid: The `renderProducts()` function dynamically generates product card HTML using template literals and appends them to the DOM. Event listeners for add-to-cart, wishlist, and quick-view are re-attached after each render cycle.

Cart System: The cart is maintained as a JavaScript array. `addToCart()` checks for existing items and increments quantity if found. `renderCart()` recalculates and re-renders the sidebar on every state change.

Wishlist: The wishlist mirrors the cart architecture without quantities. `toggleWishlist()` adds or removes items and syncs the heart icon state on all product cards simultaneously.

4.3 AI Chatbot Design

The AI shopping assistant uses a keyword-matching rule engine with the following decision tree:

- **Greetings** — Detects common greeting words and returns a personalised welcome.
- **Budget Queries** — Regular expressions extract numeric budget values and filter the catalogue.
- **Category Queries** — Keywords map to fashion, electronics, or lifestyle product lists.
- **Trend Queries** — Returns top three products on keywords like 'trending' or 'popular'.
- **Gift Queries** — Returns gift-appropriate recommendations on 'gift', 'birthday', 'surprise'.
- **Product Tags** — Each product has a tag array; matching any tag returns that product's detail.
- **Fallback** — A randomised fallback response is returned when no pattern is matched.

4.4 Novelty

The primary novelty of this project lies in the zero-cost, zero-latency AI chatbot that operates entirely client-side. Unlike cloud-dependent chatbots, this implementation has no running cost and no external API dependency. The engine is API-swappable: replacing the `getSmartReply()` function with an API call upgrades the chatbot to a full LLM without changing any other code.

The consistent purple gradient design language applied uniformly across all UI components — login modal, navbar, hero section, product cards, sidebars, and chatbot panel — creates a cohesive brand identity that distinguishes this project from typical academic submissions.

CHAPTER-5

EXPERIMENTAL RESULTS AND DISCUSSION

5.1 Testing Methodology

The application was tested manually across multiple browsers (Google Chrome 121, Mozilla Firefox 123, Microsoft Edge 121) and screen sizes (1920×1080 desktop, 1366×768 laptop, 375×667 mobile). Functional testing covered all interactive features.

5.2 Feature Test Results

Login Modal: Pass
Skip Login: Pass
Product Render: Pass
Search Filter: Pass
Sort Low–High: Pass
Add to Cart: Pass
Cart Quantity +/-: Pass
Remove from Cart: Pass
Wishlist Toggle: Pass
Quick View Modal: Pass
Chatbot Greeting: Pass
Chatbot Budget: Pass
Chatbot Trending: Pass
Checkout Demo: Pass

5.3 Performance

The application loads in under 1 second on a standard broadband connection. The JavaScript bundle (script.js) is approximately 18KB unminified and the stylesheet (style.css) is approximately 12KB. No build tools, bundlers, or package managers are required for deployment.

5.4 Discussion

All 14 tested features passed without errors across all three browsers. The responsive layout performed correctly on mobile viewports with the navbar collapsing and product cards reflowing to a single column. The chatbot correctly handled all tested query types including budget filtering, category queries, gift suggestions, and greetings.

A key limitation is the static product catalogue. Real-world deployment would require a backend API for dynamic product data and inventory management. The checkout flow is a demonstration only and does not integrate with a payment gateway.

CHAPTER-6

CONCLUSIONS AND FUTURE DIRECTIONS

6.1 Conclusions

This project successfully demonstrates the development of a feature-rich, AI-assisted e-commerce web application using only client-side technologies. All core e-commerce features — product browsing, search, cart management, wishlist, quick-view, and checkout — are implemented in a clean, responsive, and visually cohesive interface.

The AI personal shopping assistant, built as a lightweight rule-based engine, provides meaningful, context-aware recommendations without any external API dependency, ensuring zero running cost and zero latency. The project demonstrates practical application of front-end development principles within a purely client-side constraint.

6.2 Future Directions

- **Backend Integration** — Connect to a Node.js/Express or Django REST API for real-time product data, user accounts, and order management.
- **Payment Gateway** — Integrate Razorpay or Stripe for real payment processing.
- **LLM-Powered Chatbot** — Replace the rule-based engine with OpenAI GPT-4 or Anthropic Claude API calls.
- **Product Recommendations** — Implement collaborative filtering for personalised suggestions.
- **Admin Dashboard** — Build a panel for managing products, orders, and users.
- **PWA Support** — Convert the application into a Progressive Web App for offline access.
- **Multilingual Support** — Add Hindi and regional language support for Indian users.

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